

TEST BANK

CHAPTER 8

ESTUARIES AND SALT MARSHES

Matching

In this section, you are to select from the choices listed in column B the term or phrase that best matches the term or phrase listed in column A and then to place the letter for the selected choice from column B in the space provided to the left of the corresponding item in column A.

[Note to instructors: Here, the choices are correctly matched and thus should be randomly assorted before the exam is prepared.]

Column A		Column B
___ 1. fjords	_____→	A. estuaries carved by glaciers
___ 2. lagoons	_____→	B. estuaries formed by sandbars
___ 3. coastal-plain estuaries	_____→	C. formed by rising sea levels
___ 4. negative estuaries	_____→	D. have high evaporation rates
___ 5. salt-wedge estuaries	_____→	E. highest salinity is at the bottom
___ 6. stenohaline	_____→	F. intolerant of changes in salinity
___ 7. euryhaline	_____→	G. tolerant of changes in salinity
___ 8. American-type estuaries	_____→	H. dominated by emergent salt-marsh plants
___ 9. European-type estuaries	_____→	I. dominated by barren mud flats
___ 10. osmoregulation	_____→	J. control over the concentration of internal salts

Multiple Choice

Circle the choice that best completes or answers the given questions.

[N.B.: The correct answer to each of the questions is indicated by an asterisk. For most of the questions, the letter “a” is the correct choice; reorder the choices as you see fit.]

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| <p>1. Which of the following bodies of water would be classified as an estuary?</p> <ul style="list-style-type: none"> a. San Francisco Bay * b. Caspian Sea c. Aral Sea d. Great Salt Lake e. Mediterranean Sea | <p>2. Which of the following estuaries is a coastal-plain estuary?</p> <ul style="list-style-type: none"> a. Chesapeake Bay * b. San Francisco Bay c. Naeroy Fjord d. Caspian Sea e. Aral Sea |
|---|--|

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3. Which one of the following is an example of a tectonic estuary?
 - a. San Francisco Bay *
 - b. Chesapeake Bay
 - c. Aral Sea
 - d. Lake Tahoe
 - e. Naeroy Fjord
4. An isohaline is _____.
 - a. a line on a map or graph connecting regions of equal salinity *
 - b. a region where ice and salt interact
 - c. the depth at which ice forms in the sea
 - d. the depth at which salt crystallizes from seawater
 - e. None of the above
5. One organism that lives in riverine-estuary environments and then migrates to the open ocean to release its larvae is the _____.
 - a. blue crab, *Callinectes sapidus* *
 - b. polychaete worm, *Nereis diversicolor*
 - c. alga *Enteromorpha intestinalis*
 - d. cordgrass, *Spartina*
 - e. None of the above
6. When sandbars help to form an estuary by cutting off a shallow bay, the resulting estuary is called a _____ estuary.
 - a. semienclosed-bay *
 - b. coastal-plain
 - c. tectonic
 - d. fjord
 - e. neutral
7. Estuaries where freshwater input is low and evaporation is high are known as _____ estuaries.
 - a. negative *
 - b. neutral
 - c. salt-wedge
 - d. positive
 - e. marine-dominated
8. Estuaries open to the sea only during a marked wet period are called _____.
 - a. seasonal estuaries *
 - b. negative estuaries
 - c. positive estuaries
 - d. river-dominated estuaries
 - e. neutral estuaries
9. In the Chesapeake Bay estuary, surface isohalines are _____.
 - a. slanted in a way which indicates that the marine water moves farther up the eastern shore *
 - b. slanted in a way which indicates that the marine water moves farther up the western shore
 - c. even, indicating that the marine water moves up the bay equally
10. Primary productivity of salt marshes seems to be limited by which one of the following nutrients?
 - a. nitrogen *
 - b. carbon
 - c. phosphorus
 - d. iron
11. The amount of time it takes for a given volume of freshwater to discharge from an estuary is called the _____ time.
 - a. flushing *
 - b. fetch
 - c. Aeolian transport
 - d. Coriolis effect
 - e. None of the above
12. Marine organisms that are unable to tolerate much change in salinity are called _____.
 - a. stenohaline *
 - b. euryhaline
 - c. isohaline
 - d. brackish-water species
 - e. None of the above
13. Which of the following is a specialized tissue found in estuarine plants that brings oxygen to the roots?
 - a. aerenchyma *
 - b. isohaline
 - c. stenohaline
 - d. euryhaline
 - e. None of the above
14. The _____ estuary is dominated by extensive mud flats with few plants, but large populations of benthic diatoms.
 - a. European-type *
 - b. American-type
 - c. salt-wedge
 - d. neutral
 - e. river-dominated

15. The ____ estuary is dominated by extensive stands of the seagrass *Spartina*. This type of estuary produces extra carbon that flows on currents to the surrounding communities.
 - a. European-type
 - b. American-type *
 - c. salt-wedge
 - d. neutral
 - e. river-dominated
16. Some salt-marsh plants possess chemical defenses of ____ compounds.
 - a. phenolic *
 - b. lipid
 - c. protein
 - d. carbohydrate
17. Halophytes are plants capable of growing in soils _____.
 - a. of high salt content *
 - b. of low salt content
 - c. that are never covered by salt water
18. An estuary is which of the following?
 - a. A coastal embayment where freshwater and seawater mix *
 - b. A body of marine water
 - c. A coastal embayment of freshwater
 - d. An inland body of very salty water
 - e. A coastal embayment of very salty water
19. The ability to control the concentration of salts or water in internal body fluids is called _____.
 - a. osmoregulation *
 - b. aerenchyma
 - c. the Coriolis effect
 - d. isohaline
 - e. Aeolian transport
20. To control the osmotic concentration of their internal fluids, certain crustaceans use _____.
 - a. free amino acids *
 - b. salt wedges
 - c. red tides
 - d. high evaporation rates
 - e. None of the above
21. Which one of the following is not one of the four faunal types in estuaries?
 - a. terrestrial *
 - b. marine
 - c. freshwater
 - d. estuarine
 - e. transitional
22. Plants rid themselves of excessive salt by using a variety of mechanisms. Which one of the following is not one of these mechanisms?
 - a. Pumping salts out of the roots *
 - b. Salt-secreting glands
 - c. Deposition of salt on external surfaces
 - d. Shedding of leaves loaded with excess salt
23. An intermittent estuary is _____.
 - a. an estuary formed in areas where there is a marked wet and dry season *
 - b. also called a river-dominated or stratified estuary
 - c. an estuary where freshwater inflow gives similar salinities from surface to bottom at any point
 - d. an estuary where freshwater input is small and the rate of evaporation is high
24. Based on geology and geomorphology, there are four basic estuarine types. Which of the following is not an estuary type?
 - a. Fjord
 - b. Coastal plain
 - c. Tectonic
 - d. Semienclosed bay or lagoon
 - e. All of the above are estuary types *

True/False

Mark each of the given statements with the letter *T* if it is true and *F* if it is false.

[Instructors: In some cases, a few options are included for the same question.]

1. [False] If an estuary has the highest salinity near the top and the lowest at the bottom, it is called a positive estuary.
2. [True] Fiddler crabs help oxygenate anoxic sediments in the salt marsh.

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3. [True] If an estuary has the highest salinity near the bottom and the lowest at the top, it is called a river-dominated estuary.
4. [False] If an estuary has the highest salinity near the bottom and the lowest at the top, it is called a marine-dominated estuary.
5. [True] Benthic diatoms are more abundant in estuaries than planktonic diatoms.
6. [False] Planktonic diatoms are more abundant in estuaries than benthic diatoms.
7. [True] Aerenchyma is a specialized tissue present in estuarine plants that brings oxygen to the roots.
8. [True] Osmoregulators are marine organisms that are able to control their internal salt balance.
9. [False] Osmoconformers are marine organisms that are able to control their internal salt balance.
10. [True] Salt-marsh communities of grasses, shrubs, and herbs develop in soils that are inundated and drained by tides.
11. [True] Stenohaline organisms are unable to tolerate wide swings in salinity.
12. [False] Euryhaline organisms are not able to tolerate wide swings in salinity.
13. [False] The temperature range decreases as one progresses further up the estuary.
14. [True] Organisms that are truly estuarine are able to tolerate salinities between 5 and 18 psu.
15. [False] The net flow of water through a semipermeable membrane separating two salt solutions of unequal concentrations is from the higher salt concentration to the lower salt concentration.
16. [True] Diatoms dominate the phytoplankton in estuaries.
17. [False] Rotifers dominate the zooplankton of estuaries.
18. [False] The number of species inhabiting estuaries is significantly higher than the number inhabiting nearby freshwater or marine habitats.
19. [True] Halophytes grow on soils with high salt content.
20. [False] Wave action in estuaries is as important as in the rocky intertidal zone.

Short Answer (Fill in the Blank)

1. An estuary is a coastal embayment where freshwater and seawater mix.
2. Coastal-plain estuaries are estuaries that were formed when the rising sea level inundated low-lying shores after the end of the last glaciation.
3. Tectonic estuaries are estuaries formed when coastal land subsides due to geologic forces.
4. Coastal valleys deepened by glacial action that have then been inundated by the sea are called fjords.
5. Lines on a map or graph that connect regions of equal salinity are known as isohalines.
6. Seasonal estuaries have a significant wet period when freshwater inputs open the estuary to the sea.
7. In a positive or salt-wedge estuary, the salinity of the surface water decreases as one moves from the ocean toward the river.
8. In a negative estuary, the salinity of the surface water increases as one moves from the ocean toward the land.
9. Stenohaline marine organisms are unable to tolerate much change in salinity.
10. Euryhaline marine organisms are able to tolerate wide swings in salinity.
11. In estuarine plants, a specialized tissue called aerenchyma brings oxygen to the roots.
12. Certain crustaceans are able to regulate their internal salt content by changing the concentration of free amino acids. This process is called intracellular osmotic regulation.
13. Callinectes sapidus (blue crab) is an arthropod that is a commercially important shellfish along the Atlantic and Gulf coasts of the United States.
14. Tall Spartina grows high in the intertidal zone; it is inundated only once a month.
15. Organic matter produced at one location and transported to another site is termed allochthonous.

16. Flushing time is the amount of time required for a given mass of freshwater to be discharged from the estuary.
17. In the northern hemisphere, the Coriolis effect influences the flow of tidal water into a riverine–estuary system by deflecting the salt water to the right.
18. Morphological adaptations of estuarine plants include a special air-handling tissue called aerenchyma.
19. The marsh plant *Spartina alterniflora* is found on the seaward edge of New England and Southern Atlantic marshes.
20. Freshwater species are unable to tolerate salinities above 5 psu.
21. The time required for a given mass of freshwater to be discharged from the estuary is the flushing time.
22. *Spartina alterniflora* forms a mutualistic association with both mussels and fiddler crabs.

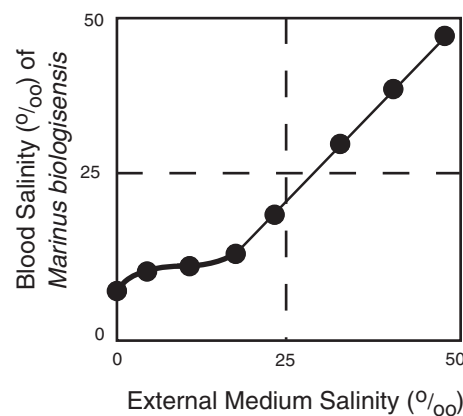
Essay Discussion Questions

Develop complete answers to these questions.

1. What sorts of catastrophic events influence estuaries, and how do they affect the estuaries? Discuss.
2. Discuss why benthic diatoms are more abundant in estuaries than planktonic diatoms.
3. In what ways can osmoregulators control their internal salt content?
4. Of what significance to estuarine crustaceans is the physiological process called intracellular osmotic regulation?
5. Using x – y coordinate axes, plot internal (body fluid) salinity of a hypothetical invertebrate vs. external salinity for an organism that is (a) an osmoregulator and (b) an osmoconformer.
6. Discuss the mechanisms that vascular plants use in saline environments to rid themselves of excess salt.
7. Discuss the behavioral mechanisms that estuarine animals use to avoid being exposed to excessive salinity.
8. Explain why salt marshes tend to be species poor.
9. What function can the open areas on pans and shallow pools serve in a salt marsh?
10. Compare and contrast the zonation patterns of two salt marshes, one that develops on a protected, low-energy beach and another that develops on an open, moderate-energy beach.
11. What practical problems do researchers encounter when they attempt to study the primary production of vascular, emergent plants in the salt marsh?
12. Draw a typical oxygen profile depicting oxygen content in the water column of an estuary in which there are inputs that contain raw sewage and industrial wastes. In essay form, provide support for your figure.
13. Bacteria are very important in the intertidal community. Name two different roles that bacteria play in the sand and mud of the intertidal shore line.
14. Sessile, intertidal organisms of the rocky shore and similar sessile organisms that inhabit the surfaces of pilings often show marked variations in the age structure of the populations such that a single age class dominates. How can such a skewed age structure become established?
15. Describe a salt-wedge estuary. Why is it given this unusual name?
16. What is an estuary? Why are bodies such as the Great Salt Lake (Utah) and Mono Lake (California) not considered estuaries?
17. Distinguish among the origins and geomorphologies of the following estuaries: coastal-plain estuary, tectonic estuary, semienclosed bay (lagoon), and fjord.
18. Assume there is a river estuary with two towns on opposite sides of the river (for example, as is found in the area of the Chesapeake Bay; see Figure 8.4 in the text). As one proceeds down the river, the people living in the town on the right-hand side can catch freshwater fish from the bank in areas much closer to the sea than those on the left-hand side. However, those people who live on the left-hand side can catch their own blue crabs and gather oysters and clams much further from the ocean. Explain why. *[Instructors must supply the figure.]*
19. The salt marsh is a stressful environment. What strategies have animals evolved to minimize these stresses?

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20. What morphological adaptations have estuarine plants evolved to live in an environment of high salt and anoxic sediments?
21. How is the fiddler crab, *Uca*, like an earthworm? Discuss.
22. Why is the keystone-species concept not particularly useful when considering a salt marsh?
23. How does decomposition of salt-marsh plants increase the plants' food value to animals?
24. Where do fugitive plant species survive in the salt marsh? What role do such plants play in these habitats?
25. How does *Juncus gerardi* outcompete other marsh plants in New England marshes?
26. Pick a North American salt marsh and describe its zonation pattern.
27. What terrestrial animals represent major components of the fauna of the salt marsh?
28. How does bill length in European shorebirds demonstrate the concept of resource partitioning? Discuss.
29. If estuaries possess a relatively abundant supply of nutrients, why do the phytoplankton in these habitats have a low annual rate of production?
30. What factors affect production in a salt marsh?
31. Discuss the factors that explain the migration of the marsh periwinkle (*Littorina irrorata*) on cordgrass blades. How could you develop one or more experimental tests of your ideas?
32. How does a mixed-sediment suspension (in terms of particle size) settle out in a still column of water? Now consider the same question for water that flows as in a stream which enters a lake or an estuary.
33. Draw a typical oxygen profile depicting oxygen content in the water column of an estuary. How does runoff from terrestrial ecosystems influence the oxygen profile? Assume two different conditions to answer this question: The terrestrial input is (a) fine sand and silt or (b) industrial wastes.
34. Why are the estuarine phytoplankton a fairly depauperate group?
35. Why are most marine invertebrates osmoconformers? How does Figure 8.9 show that *Nereis diversicolor* is an osmoregulator? Why is osmoregulatory ability found in several taxa of more advanced estuarine animals? [Instructors must supply the figure.]
36. Why is *Nereis diversicolor* an osmoconformer as salinity decreases from 32 psu (or increases from that level), but an osmoregulator when the concentration reaches about 15 psu or less? Discuss.
37. In general, bivalve mollusks are poor osmoregulators, at least in terms of physiological ability to regulate their salt concentration. What simple behaviors do mollusks exhibit to regulate their salt concentration?
38. How do physical and biological factors interact to give zonal patterns in New England salt marshes?
39. What are the major factors that limit algal primary production in an estuary?
40. The given figure shows the results of an experiment on osmoregulation in a hypothetical marine worm: *Marinus biologisensis*. Is this invertebrate an osmoconformer or an osmoregulator? Defend your answer.



41. What is meant by consumer control of an ecosystem, and how does this concept relate to the dynamics of salt marshes?
42. Discuss how shoreline development influences the process of eutrophication in estuaries.
43. Discuss the trophic cascade model as it relates to salt-marsh ecosystems.
44. How have snow geese influenced the marsh ecosystems of Hudson Bay?
45. What is peat, how does it develop, and of what importance is it to marsh development?
46. How can peat act as a buffer to short-term environmental changes?
47. Discuss the competitive interactions of planktonic diatoms and sea grasses in the Chesapeake Bay.
48. What is a salt pan, and how do such places develop?
49. Challenge Question: Explore the concept that ecosystems provide services (known as ecosystem services) to the people who live near them and even to people who live quite far away.
50. Challenge Question: What ecosystem services do marshes provide? Discuss the relative importance of each.
51. Challenge Question: Discuss nutrient limitation in vascular plants that live in terrestrial habitats vs. those that live in salt marshes.
52. Challenge Question: What sorts of similar behaviors might you expect to see in those animals which inhabit deserts and those which inhabit the intertidal environment (either the rocky intertidal zone or the estuarine salt marsh)?

